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Using logistic regression procedures for estimating receiver operating characteristic curves

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Summary

Estimation of a receiver operating characteristic, ROC, curve is usually based either on a fully parametric model such as a normal model or on a fully nonparametric model. In this paper, we explore a semiparametric approach by assuming a density ratio model for disease and disease-free densities. This model has a natural connection with the logistic regression model. The proposed semiparametric approach is more robust than a fully parametric approach and is more efficient than a fully nonparametric approach. Two real examples demonstrate that the ROC curve estimated by our semiparametric method is much smoother than that estimated by the nonparametric method.

Some key words: Bootstrap; Density ratio model; Diagnostic test; Gaussian process; Logistic regression model; ROC curve; Semiparametric likelihood; Sensitivity; Specificity.

1. Introduction

The accuracy of a medical diagnostic test is typically summarised by its sensitivity and specificity, defined respectively as the probability that a truly diseased subject has a positive test result and the probability that a truly nondiseased subject has a negative test result. When the sampling distribution of a diagnostic variable is continuous, one commonly uses a receiver operating characteristic, ROC, curve, defined as a plot of the sensitivity versus one minus the specificity across all possible cut-points of the variable. For comprehensive reviews of recent developments in ROC curves see for example Begg (1991), Hsieh & Turnbull (1996), Pepe (2000a) and Zhou et al. (2002). In the literature, estimation of an ROC curve may be based either on a parametric model or on a fully nonparametric model. A correctly specified parametric approach usually yields a smoother and more accurate ROC curve than does the nonparametric approach. The parametric approach may, however, suffer from serious model misspecification bias if a parametric model is incorrectly specified. Goddard & Hinberg (1990) found that the normal assumption is not a good choice for describing the distributions of serum prostate acid phosphatase, a potential biomarker for prostate cancer. By contrast, the nonparametric analysis of an

ROC curve is a robust method, but the resulting ROC curve is not smooth, especially for small sample sizes.

As a compromise between the parametric and nonparametric approaches, the binormal model is a popular semiparametric approach to modelling an ROC curve. In this model, the test results from diseased and nondiseased populations, after a possible unknown monotonic transformation, are assumed to have normal distributions. Hsieh & Turnbull (1996) established asymptotic theory for the nonparametric ROC-curve estimator and proposed a generalised least-squares procedure for estimating parameters under the binormal model. Metz et al. (1998) proposed a rank-based method for estimating an ROC curve under the binormal model by grouping the observed data on the basis of the 'truth-state runs' in rank-ordered test results. For the binormal model, Zou & Hall (2000) employed a rank-based likelihood to estimate parameters in the ROC curve. By interpreting each point on the ROC curve as being a conditional expectation of a binary random variable, Pepe (2000b) used generalised linear model methods for binary data to model the ROC curve semiparametrically and to estimate the underlying parameters. Li et al. (1999) considered a semiparametric approach to estimation of an ROC curve by modelling the test results from nondiseased and diseased populations nonparametrically and parametrically, respectively.

Let $X_1, \ldots, X_{n_0}$ denote independent and identically distributed test results from a nondiseased population and, independently of the X_i, let $Y_1, \ldots, Y_{n_1}$ be independent and identically distributed test results from a diseased population. If G and F represent respectively the distribution functions of X_1 and Y_1, the ROC curve is a plot of

$$R(s) = 1 - F\{G^{-1}(1-s)\}$$

against s, for $s \in [0, 1]$. The area under the ROC curve is

$$A = \int_0^1 R(s)\, ds = \text{pr}(Y_1 > X_1).$$

In this paper, we consider an alternative semiparametric approach to the estimation of $R(s)$. Instead of modelling directly the distribution functions F and G of test results for a given disease status, one may model the probability of the disease status for a given test result. Logistic regression models are commonly used for such modelling (Anderson, 1972), but the connection between logistic regression and ROC curves is unclear. The SAS PROC LOGISTIC code (SAS/STAT Software, 1997, Ch. 16), designed specifically to fit logistic regression models, has an option that produces an empirical or nonparametrically estimated ROC curve based on the empirical distribution functions of the nondiseased sample $(X_1, \ldots, X_{n_0})$ and the diseased sample $(Y_1, \ldots, Y_{n_1})$. This nonparametrically estimated ROC curve ignores the information provided by the fitted logistic regression model even if this fits the data well. The present paper explores the connection between ROC curves and logistic regression models and provides a method that fills the gap between the parametric and nonparametric approaches to the estimation of ROC curves.

Let $D = 1$ or $D = 0$ denote the disease or nondisease status. For a given test result $X = x$, the logistic regression model is

$$\text{pr}(D = 1 \mid X = x) = \frac{\exp\{\alpha^* + \beta^{\mathrm{T}} r(x)\}}{1 + \exp\{\alpha^* + \beta^{\mathrm{T}} r(x)\}}, \qquad (1{\cdot}1)$$

where α^* is a scalar parameter, β is a $p \times 1$ vector parameter, and $r(x)$ is a $p \times 1$ smooth vector function of x. Then $F(x) = \text{pr}(X \leqslant x \mid D = 1)$ and $G(x) = \text{pr}(X \leqslant x \mid D = 0)$. Let $f(x)$

and $g(x)$ be the density functions corresponding to $F(x)$ and $G(x)$. Qin & Zhang (1997) showed that

$$\frac{f(x)}{g(x)} = \exp\{\alpha + \beta^{\mathrm{T}} r(x)\},$$

where $\alpha = \alpha^* + \log[\{1 - \mathrm{pr}(D=1)\}/\mathrm{pr}(D=1)]$. Hence, we arrive at the following two-sample semiparametric density ratio model in which the two unknown density functions f and g are linked by an 'exponential tilt' $\exp\{\alpha + \beta^{\mathrm{T}} r(x)\}$:

$$\begin{aligned} X_1, \ldots, X_{n_0} &\text{ are independent with density } g(x), \\ Y_1, \ldots, Y_{n_1} &\text{ are independent with density } f(x) = \exp\{\alpha + \beta^{\mathrm{T}} r(x)\} g(x). \end{aligned} \tag{1·2}$$

The relationship between the density ratio function, also known as the likelihood ratio function, and the risk function $\mathrm{pr}(D=1 \mid X)$ is well known in the Bayesian literature; see for example McLachlan (1992, p. 255). Model (1·1) or (1·2) is satisfied with $r(x) = x$ if F and G are each normal with a common variance but different means; if the variances differ then a quadratic term is needed so that $r(x) = (x, x^2)^{\mathrm{T}}$ in model (1·1) or (1·2). Kay & Little (1987) discussed density ratio models for some conventional distributions. One notable feature of estimating $R(s)$ under model (1·2) is that we model G nonparametrically and F semiparametrically. We anticipate that, for estimating an ROC curve, the logistic regression model based approach will be more robust than the normal distribution based approach and more efficient than a fully nonparametric approach.

Green & Swets (1966, p. 36) pointed out that the slope of the ROC curve is the likelihood ratio function. Eguchi & Copas (2002) and McIntosh & Pepe (2002) presented this property under the logistic regression model (1·1). By differentiating $R(s)$ with respect to s under the density ratio model (1·2), we have

$$R'(s) = \frac{dR(s)}{ds} = \frac{f\{G^{-1}(1-s)\}}{g\{G^{-1}(1-s)\}} = \exp[\alpha + \beta^{\mathrm{T}} r\{G^{-1}(1-s)\}], \tag{1·3}$$

so that

$$R(s) = \int_0^s R'(t)\, dt = \int_0^s \exp[\alpha + \beta^{\mathrm{T}} r\{G^{-1}(1-t)\}]\, dt \equiv \psi(\alpha, \beta, G).$$

Thus, under model (1·2), the slope of the ROC curve $R(s)$ is the semiparametric likelihood ratio function or the semiparametric exponential tilt $\exp[\alpha + \beta^{\mathrm{T}} r\{G^{-1}(1-s)\}]$. Furthermore, the density ratio model (1·2) implies that the ROC curve $R(s)$ is a known function of (α, β) and the baseline distribution function G which is in general unknown. This contrasts with the classical binormal model and the generalised linear model based ROC regression model of Pepe (2000b) in which the ROC curve is modelled parametrically with some specified link function and known basis functions. A nice property of the ROC curve is its invariance under monotone increasing transformations of the diagnostic variable. Although the density ratio model (1·2) does not have this property, it is semi-invariant in the sense that the two transformed samples, $m(X_1), \ldots, m(X_{n_0})$ and $m(Y_1), \ldots, m(Y_{n_1})$, using a monotone increasing function $m(.)$, have density ratio $\exp[\alpha + \beta^{\mathrm{T}} r\{m^{-1}(x)\}]$ with $m^{-1}(.)$ being the inverse function of $m(.)$, the same form as the original except for $m^{-1}(x)$ in place of x.

In § 2 we propose our semiparametric estimators of the ROC curve $R(s)$ and the ROC-curve area A under the density ratio model (1·2). We also present in § 2 some asymptotic results for the proposed semiparametric estimators and a bootstrap method for constructing confidence bands for the ROC curve. In § 3, we discuss asymptotic efficiency of the proposed semiparametric ROC-curve estimator and report results of simulation studies. In § 4, we apply our proposed semiparametric estimators to two real examples. Proofs are provided in the Appendix.

2. MAIN RESULTS

Let $\{T_1, \ldots, T_n\}$ denote the combined test results $\{X_1, \ldots, X_{n_0}; Y_1, \ldots, Y_{n_1}\}$ with $n = n_0 + n_1$. Let $\{t_1, \ldots, t_n\}$ represent the observed values of $\{T_1, \ldots, T_n\}$. Following Qin & Zhang (1997), let

$$\tilde{F}(t) = \frac{1}{n_0} \sum_{i=1}^{n} \frac{\exp\{\tilde{\alpha} + \tilde{\beta}^{\mathrm{T}} r(T_i)\} I(T_i \leqslant t)}{1 + \rho \exp\{\tilde{\alpha} + \tilde{\beta}^{\mathrm{T}} r(T_i)\}}, \quad \tilde{G}(t) = \frac{1}{n_0} \sum_{i=1}^{n} \frac{I(T_i \leqslant t)}{1 + \rho \exp\{\tilde{\alpha} + \tilde{\beta}^{\mathrm{T}} r(T_i)\}},$$

where $\rho = n_1/n_0$ and $(\tilde{\alpha}, \tilde{\beta})$ is the solution of the score equations

$$\frac{\partial l(\alpha, \beta)}{\partial \alpha} = n_1 - \sum_{i=1}^{n} \frac{\rho \exp\{\alpha + \beta^{\mathrm{T}} r(t_i)\}}{1 + \rho \exp\{\alpha + \beta^{\mathrm{T}} r(t_i)\}} = 0,$$

$$\frac{\partial l(\alpha, \beta)}{\partial \beta} = \sum_{j=1}^{n_1} r(y_j) - \sum_{i=1}^{n} \frac{\rho \exp\{\alpha + \beta^{\mathrm{T}} r(t_i)\}}{1 + \rho \exp\{\alpha + \beta^{\mathrm{T}} r(t_i)\}} r(t_i) = 0. \tag{2·1}$$

As shown in Qin & Zhang (1997), $(\tilde{\alpha}, \tilde{\beta}, \tilde{G})$ is the set of maximum semiparametric likelihood estimators of (α, β, G) in the sense that $(\tilde{\alpha}, \tilde{\beta}, \tilde{G})$ maximises the full likelihood function

$$L(\alpha, \beta, G) = \prod_{i=1}^{n_0} dG(x_i) \prod_{j=1}^{n_1} \exp\{\alpha + \beta^{\mathrm{T}} r(y_j)\}\, dG(y_j).$$

On the basis of $\tilde{F}(t)$ and $\tilde{G}(t)$, we propose under model (1·2) to estimate the ROC curve $R(s)$ by

$$\tilde{R}(s) = 1 - \tilde{F}\{\tilde{G}^{-1}(1-s)\}, \quad s \in [0, 1],$$

and the area A under $R(s)$ by

$$\tilde{A} = \int_0^1 \tilde{R}(s)\, ds.$$

We call $\tilde{R}(s)$ and $\tilde{A}$ the maximum semiparametric likelihood estimators of $R(s)$ and A, respectively. By (1·3), the slope of the proposed ROC curve estimator $\tilde{R}(s)$ is the estimated likelihood ratio function.

Let $\hat{F}(x) = n_1^{-1} \sum_{i=1}^{n_1} I(Y_i \leqslant x)$ and $\hat{G}(x) = n_0^{-1} \sum_{i=1}^{n_0} I(X_i \leqslant x)$ be the nonparametric maximum likelihood estimators of F and G. Then the nonparametric maximum likelihood estimators of $R(s)$ and A are $\hat{R}(s) = 1 - \hat{F}\{\hat{G}^{-1}(1-s)\}$ and $\hat{A} = \int_0^1 \hat{R}(s)\, ds$, respectively. Since $\tilde{F}$ and $\tilde{G}$ have a larger support $\{t_1, \ldots, t_n\}$ than either $\hat{F}$ or $\hat{G}$, the semiparametrically estimated ROC curve $\tilde{R}(s)$ has more and smaller jumps than $\hat{R}(s)$.

Let (α_0, β_0) be the true value of (α, β) under model (1·2). The following theorem whose proof is given in the Appendix establishes the weak convergence of $\tilde{R}(s)$.

THEOREM 1. *Let $0 < a < b < 1$ be given. Suppose that F and G have continuous positive densities f and g, respectively, on $[G^{-1}(a) - \varepsilon, G^{-1}(b) + \varepsilon]$ for some $\varepsilon > 0$. If A is positive definite, then under model* (1·2) *we have $\sqrt{n}\{\tilde{R}(s) - R(s)\} \to W\{G^{-1}(1-s)\}$ weakly in*

$\mathscr{D}[1-b, 1-a]$, where $\mathscr{D}[1-b, 1-a]$ denotes the set of all real-valued functions of $[1-b, 1-a]$ that are right continuous and possess left-hand limits at each point and W is a Gaussian process with mean 0 *and covariance function $E\{W(s)W(t)\} = K(s, t)$ given by*

$$K(s,t) = \frac{1+\rho}{\rho}\{F(s\wedge t) - F(s)F(t)\} + (1+\rho)\exp\{\alpha_0 + \beta_0^{\mathrm{T}} r(s)\}\exp\{\alpha_0 + \beta_0^{\mathrm{T}} r(t)\}$$
$$\times \{G(s\wedge t) - G(s)G(t)\} - \frac{1+\rho}{\rho}[1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}} r(s)\}]$$
$$\times [1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}} r(t)\}]\left\{A_0(s\wedge t) - (A_0(s), A_1^{\mathrm{T}}(s))A^{-1}\begin{pmatrix} A_0(t) \\ A_1(t)\end{pmatrix}\right\}$$

for $-\infty \leqslant s, t \leqslant \infty$. Here, $A_0(t)$, $A_1(t)$ and A are defined in (A·1) *of the Appendix.*

Theorem 1 immediately implies that

$$\sqrt{n}\{\tilde{R}(s) - R(s)\} \to N[0, K\{G^{-1}(1-s), G^{-1}(1-s)\}]$$

in distribution for each fixed $s \in [1-b, 1-a]$. Let $\tilde{K}(s, t)$ be an empirical version of $K(s, t)$ with (α_0, β_0, G) replaced by $(\tilde{\alpha}, \tilde{\beta}, \tilde{G})$. Then an approximate level $1-\alpha$ pointwise confidence interval for $R(s)$ is given by

$$\tilde{R}(s) \pm z_{1-\alpha/2}[\tilde{K}\{\tilde{G}^{-1}(1-s), \tilde{G}^{-1}(1-s)\}/n]^{\frac{1}{2}}$$

for $s \in [1-b, 1-a]$, where z_α is such that $\mathrm{pr}(Z \leqslant z_\alpha) = \alpha$ with $Z \sim N(0, 1)$.

Another immediate consequence of Theorem 1 is the following result regarding the asymptotic distribution of $\tilde{A}$.

Theorem 2. *Under the conditions of Theorem* 1, *we have*

$$\sqrt{n}(\tilde{A} - A) \to N\{0, \sigma^2(\alpha_0, \beta_0, G)\}$$

in distribution, where

$$\sigma^2(\alpha_0, \beta_0, G) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} K(s,t)\, dG(s)\, dG(t).$$

Let $\tilde{\sigma}^2 = \sigma^2(\tilde{\alpha}, \tilde{\beta}, \tilde{G})$ be an empirical version of $\sigma^2(\alpha_0, \beta_0, G)$ with (α_0, β_0, G) replaced by $(\tilde{\alpha}, \tilde{\beta}, \tilde{G})$. Then an approximate level $1-\alpha$ confidence interval for A is given by $\tilde{A} \pm z_{1-\alpha/2}\tilde{\sigma}/\sqrt{n}$. Since the ROC curve estimate $\tilde{R}(s)$ and the estimated area $\tilde{A}$ under the ROC curve are restricted to the range $(0, 1)$ and their values will be close to the upper boundary for good tests, the logit transform would be preferable in practice to constructing confidence intervals for $R(s)$ and A. The delta method along with Theorems 1 and 2 can be applied to find the asymptotic standard errors of $\mathrm{logit}\{\tilde{R}(s)\}$ and $\mathrm{logit}(\tilde{A})$.

The covariance function $K(s, t)$ in Theorem 1 and the asymptotic variance $\sigma^2(\alpha_0, \beta_0, G)$ in Theorem 2 depend on the density functions $f(x)$ and $g(x)$ only through their ratio $f(x)/g(x) = \exp\{\alpha_0 + \beta_0^{\mathrm{T}} r(x)\}$, so that we do not need to estimate $f(x)$ or $g(x)$ when we seek estimators for $K(s, t)$ and $\sigma^2(\alpha_0, \beta_0, G)$. This is in contrast to the semiparametric approach of Li et al. (1999) in which a kernel density estimator of $g(x)$ is needed for estimating the covariance function and asymptotic variance in their Proposition 1 and Corollary 1.

Since no analytic expression appears to be available for the distribution function of

$$\sup_{1-b\leqslant s\leqslant 1-a} |W\{G^{-1}(1-s)\}|,$$

we present a bootstrap method for constructing confidence bands for the ROC curve $R(s)$. Let $X_1^*, \ldots, X_{n_0}^*$ be a random sample from $\tilde{G}$ and, independently of the X_i^*, let $Y_1^*, \ldots, Y_{n_1}^*$ be a random sample from $\tilde{F}$. Furthermore, let $T_1^*, \ldots, T_n^*$ be the combined bootstrap sample with observed values denoted by $t_1^*, \ldots, t_n^*$ and let $(\tilde{\alpha}^*, \tilde{\beta}^*)$ be the solution of the score equations in (2·1) with the t_i replaced by the t_i^*. Moreover, let

$$\tilde{F}^*(t) = \frac{1}{n_0} \sum_{i=1}^{n} \frac{\exp\{\tilde{\alpha}^* + \tilde{\beta}^{*\mathrm{T}} r(T_i^*)\} I(T_i^* \leqslant t)}{1 + \rho \exp\{\tilde{\alpha}^* + \tilde{\beta}^{*\mathrm{T}} r(T_i^*)\}},$$

$$\tilde{G}^*(t) = \frac{1}{n_0} \sum_{i=1}^{n} \frac{I(T_i^* \leqslant t)}{1 + \rho \exp\{\tilde{\alpha}^* + \tilde{\beta}^{*\mathrm{T}} r(T_i^*)\}}.$$

Then the corresponding bootstrap versions of $\tilde{R}(s)$ and $\tilde{A}$ are

$$\tilde{R}^*(s) = 1 - \tilde{F}^*\{\tilde{G}^{*-1}(1-s)\}, \quad \tilde{A}^* = \int_0^1 \tilde{R}^*(s)\, ds.$$

We propose to approximate the critical values of $\sup_{1-b \leqslant s \leqslant 1-a} \sqrt{n}|\tilde{R}(s) - R(s)|$ and $\tilde{A}$ by those of $\sup_{1-b \leqslant s \leqslant 1-a} \sqrt{n}|\tilde{R}^*(s) - \tilde{R}(s)|$ and $\tilde{A}^*$, respectively. In particular, if

$$r_{1-\gamma}^n = \inf\left[s : P^* \left\{ \sup_{1-b \leqslant s \leqslant -a} \sqrt{n}|\tilde{R}^*(s) - \tilde{R}(s)| \leqslant s \right\} \geqslant 1 - \gamma \right]$$

with P^* representing bootstrap probability under $\tilde{F}^*$ or $\tilde{G}^*$, then a level $1-\gamma$ bootstrap confidence band for the ROC curve $R(s)$ under model (1·2) is given by

$$\left(\tilde{R}(.) - \frac{r_{1-\gamma}^n}{\sqrt{n}}, \tilde{R}(.) + \frac{r_{1-\gamma}^n}{\sqrt{n}} \right). \tag{2·2}$$

3. Efficiency comparison

In this section we compare the maximum semiparametric likelihood estimator $\tilde{R}(s) = 1 - \tilde{F}\{\tilde{G}^{-1}(1-\mathrm{s})\}$ with the nonparametric maximum likelihood estimator $\hat{R}(s) = 1 - \hat{F}\{\hat{G}^{-1}(1-s)\}$. Hsieh & Turnbull (1996) established asymptotic theory for the nonparametric ROC-curve estimator $\hat{R}(s)$. A standard nonparametric analysis shows that, under the conditions of Theorem 1,

$$\sqrt{n}\{\hat{R}(s) - R(s)\} \to U\{G^{-1}(1-s)\}$$

weakly in $\mathscr{D}[1-b, 1-a]$, where U is a Gaussian process with mean 0 and covariance function $E\{U(s)U(t)\} = M(s, t)$ given by

$$M(s,t) = \frac{1+\rho}{\rho}\{F(s \wedge t) - F(s)F(t)\} + (1+\rho)\exp\{\alpha_0 + \beta_0^{\mathrm{T}} r(s)\}\exp\{\alpha_0 + \beta_0^{\mathrm{T}} r(t)\}$$
$$\times \{G(s \wedge t) - G(s)G(t)\}$$

for $-\infty \leqslant s, t \leqslant \infty$. This result implies that

$$\sqrt{n}\{\hat{R}(s) - R(s)\} \to N[0, M\{G^{-1}(1-s), G^{-1}(1-s)\}]$$

in distribution for each fixed $s \in [1-b, 1-a]$. Let $d(s,t) = M(s,t) - K(s,t)$. Then the asymptotic relative efficiency of $\hat{R}(s)$ with respect to $\tilde{R}(s)$ is given by

$$e(s) = \frac{K\{G^{-1}(1-s), G^{-1}(1-s)\}}{M\{G^{-1}(1-s), G^{-1}(1-s)\}} = 1 - \frac{d\{G^{-1}(1-s), G^{-1}(1-s)\}}{M\{G^{-1}(1-s), G^{-1}(1-s)\}}.$$

It is seen from Theorem 1 that

$$d(s,t)=\frac{1+\rho}{\rho}[1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(s)\}][1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}]$$
$$\times\left\{A_0(s\wedge t)-(A_0(s),A_1^{\mathrm{T}}(s))A^{-1}\begin{pmatrix}A_0(t)\\A_1(t)\end{pmatrix}\right\}.$$

According to Theorem 2.1 of Zhang (2000),

$$\frac{\rho^2 d(s,t)}{[1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(s)\}][1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}]}$$
$$=\rho(1+\rho)\left\{A_0(s\wedge t)-(A_0(s),A_1^{\mathrm{T}}(s))A^{-1}\begin{pmatrix}A_0(t)\\A_1(t)\end{pmatrix}\right\}$$

is the asymptotic covariance function of $\sqrt{n}\{\tilde{G}(t)-\hat{G}(t)\}$. Consequently, we have that

$$d\{G^{-1}(1-s),G^{-1}(1-s)\}\geqslant 0$$

and $e(s)\leqslant 1$ for $s\in[1-b,1-a]$. Therefore, $\tilde{R}(s)$ is asymptotically more efficient than $\hat{R}(s)$ for each fixed $s\in[1-b,1-a]$.

We now report a small simulation study of the finite sample performance of the proposed ROC curve estimator $\tilde{R}(s)$. In our study, we assume that $g(x)$ is the density function of a $N(\mu_0,\sigma_0^2)$ distribution and $f(x)$ is the density function of a $N(\mu_1,\sigma_1^2)$ distribution. Then model (1·2) holds with $r(x)=(x,x^2)^{\mathrm{T}}$, $\beta=(\beta_1,\beta_2)^{\mathrm{T}}$ and

$$\alpha=\log\frac{\sigma_0}{\sigma_1}+\frac{1}{2}\left(\frac{\mu_0^2}{\sigma_0^2}-\frac{\mu_1^2}{\sigma_1^2}\right),\quad \beta_1=\frac{\mu_1}{\sigma_1^2}-\frac{\mu_0}{\sigma_0^2},\quad \beta_2=\frac{1}{2}\left(\frac{1}{\sigma_0^2}-\frac{1}{\sigma_1^2}\right).$$

The corresponding ROC curve is the conventional binormal model $R(s)=\Phi\{\gamma_1+\gamma_2\Phi^{-1}(s)\}$ with $\gamma_1=(\mu_1-\mu_0)/\sigma_1$ and $\gamma_2=\sigma_0/\sigma_1$. Here $\Phi(.)$ is the standard normal distribution function. Our aim is to compare the biases and standard errors of the proposed ROC-curve estimator $\tilde{R}(s)$, the GLM ROC-curve estimator $\tilde{R}_P(s)$ of Pepe (2000b), the LABROC ROC-curve estimator $\tilde{R}_M(s)$ of Metz et al. (1998) and the nonparametric ROC-curve estimator $\hat{R}(s)$. A FORTRAN program implementing the LABROC4 algorithm was employed to calculate the LABROC ROC-curve estimator $\tilde{R}_M(s)$. It should be noted that both the estimators of Pepe (2000b) and Metz et al. (1998) yield smooth binormal ROC curves in contrast to the nonparametric and proposed semiparametric estimators.

We considered $s\in\{0{\cdot}1,0{\cdot}2,0{\cdot}3,0{\cdot}4,0{\cdot}5,0{\cdot}6,0{\cdot}7,0{\cdot}8,0{\cdot}9\}$ and sample sizes of $(n_0,n_1)=(40,60)$ and $(n_0,n_1)=(60,40)$. Furthermore, we let $\mu_0=0$, $\mu_1=2$ and $\sigma_0=\sigma_1=1$ be fixed so that $\alpha=-2$, $\beta_1=2$, $\beta_2=0$, $\gamma_1=2$ and $\gamma_2=1$. For each pair (n_0,n_1), we generated 1000 independent sets of combined random samples from the $N(\mu_0,\sigma_0^2)$ and $N(\mu_1,\sigma_1^2)$ distributions. The simulation results are summarised in Table 1.

In Table 1, bias$\{\tilde{R}(s)\}$ and SE$\{\tilde{R}(s)\}$ stand for, respectively, the average of 1000 biases of $\tilde{R}(s)$ and the sample standard error of 1000 estimates $\tilde{R}(s)$. Other notation, bias$\{\tilde{R}_P(s)\}$, SE$\{\tilde{R}_P(s)\}$, bias$\{\tilde{R}_M(s)\}$, SE$\{\tilde{R}_M(s)\}$, bias$\{\hat{R}(s)\}$ and SE$\{\hat{R}(s)\}$, is defined similarly. Table 1 reveals that $\tilde{R}(s)$ produces a smaller standard error than $\hat{R}(s)$ for each s and than $\tilde{R}_P(s)$ and $\tilde{R}_M(s)$ for most values of s. In addition, the biases of $\tilde{R}(s)$ are slightly smaller than those of $\tilde{R}_M(s)$ and are slightly larger than those of $\tilde{R}_P(s)$ and $\hat{R}(s)$ for most values of s. A further calculation indicates that $\tilde{R}(s)$ has a smaller mean squared error than $\hat{R}(s)$ for each s and than $\tilde{R}_P(s)$ and $\tilde{R}_M(s)$ for most values of s. In summary, our simulation study indicates

Table 1. *Efficiency comparison. Biases and standard errors of* $\tilde{R}(s)$, $\tilde{R}_P(s)$, $\tilde{R}_M(s)$ *and* $\hat{R}(s)$

s	bias$\{\tilde{R}(s)\}$	bias$\{\tilde{R}_P(s)\}$	bias$\{\tilde{R}_M(s)\}$	bias$\{\hat{R}(s)\}$	SE$\{\tilde{R}(s)\}$	SE$\{\tilde{R}_P(s)\}$	SE$\{\tilde{R}_M(s)\}$	SE$\{\hat{R}(s)\}$
				$n_0 = 40$, $n_1 = 60$				
0·1	0·0127	−0·0296	0·0004	0·0168	0·0805	0·0885	0·0805	0·0904
0·2	0·0083	−0·0089	−0·0006	0·0049	0·0500	0·0559	0·0487	0·0603
0·3	0·0042	−0·0024	−0·0018	0·0011	0·0337	0·0381	0·0331	0·0429
0·4	0·0016	−0·0007	−0·0027	−0·0005	0·0232	0·0264	0·0235	0·0305
0·5	−0·0001	−0·0005	−0·0031	0·0004	0·0161	0·0180	0·0168	0·0217
0·6	−0·0010	−0·0007	−0·0031	0·0006	0·0109	0·0119	0·0117	0·0152
0·7	−0·0014	−0·0008	−0·0026	−0·0012	0·0070	0·0074	0·0078	0·0115
0·8	−0·0012	−0·0007	−0·0019	−0·0007	0·0041	0·0041	0·0047	0·0073
0·9	−0·0008	−0·0004	−0·0010	−0·0005	0·0020	0·0017	0·0023	0·0042
				$n_0 = 60$, $n_1 = 40$				
0·1	0·0149	−0·0199	0·0057	0·0107	0·0785	0·0853	0·0764	0·0919
0·2	0·0089	−0·0029	0·0034	0·0022	0·0514	0·0564	0·0500	0·0645
0·3	0·0044	0·0009	0·0004	0·0013	0·0352	0·0393	0·0362	0·0456
0·4	0·0014	0·0010	−0·0017	0·0013	0·0246	0·0274	0·0266	0·0320
0·5	−0·0004	0·0002	−0·0029	−0·0016	0·0173	0·0189	0·0193	0·0253
0·6	−0·0014	−0·0004	−0·0032	−0·0010	0·0119	0·0127	0·0137	0·0181
0·7	−0·0017	−0·0008	−0·0029	−0·0005	0·0079	0·0081	0·0093	0·0126
0·8	−0·0015	−0·0008	−0·0022	−0·0004	0·0049	0·0046	0·0058	0·0081
0·9	−0·0009	−0·0005	−0·0012	−0·0002	0·0025	0·0021	0·0031	0·0046

$\tilde{R}(s)$ is the proposed semiparametric estimator of $R(s)$, $\tilde{R}_P(s)$ is Pepe's estimator of $R(s)$, $\tilde{R}_M(s)$ is the estimator of $R(s)$ of Metz et al. (1998) and $\hat{R}(s)$ is the empirical estimator of $R(s)$.

that the proposed semiparametric estimator $\tilde{R}(s)$ is comparable to $\tilde{R}_P(s)$ and $\tilde{R}_M(s)$ and is better than $\hat{R}(s)$ in terms of mean squared error.

4. Examples

Example 1. As reported in Wieand et al. (1989), sera from $n_0 = 51$ control patients with pancreatitis and $n_1 = 90$ case patients with pancreatic cancer were studied at the Mayo Clinic with a cancer antigen, CA-125, and with a carbohydrate antigen, CA19-9. By employing the GLM ROC regression model methodology with the probit link, Pepe (2000b) fitted a binormal model to this pancreatic cancer dataset and reported a good fit. Here, we apply our methodolgy to part of this dataset with test results associated only with CA19-9. A log transformation was used in our data analysis. As a test of the validity of the logistic regression model (1·1) or its equivalent semiparametric density ratio model (1·2) with $r(x) = (x, x^2)^{\mathrm{T}}$, the Kolmogorov–Smirnov-type statistic of Qin & Zhang (1997) yields the observed p-value of 0·769 based on 5000 bootstrap replications, indicating a good fit of model (1·1) or (1·2) to the pancreatic cancer dataset. The proposed semiparametric ROC-curve estimator $\tilde{R}(s)$ and the nonparametric ROC-curve estimator $\hat{R}(s)$ are plotted in Fig. 1(a) along with the 95% bootstrap confidence band in (2·2). Figure 1(a) indicates that the model-based estimator $\tilde{R}(s)$ is much smoother than the nonparametric estimator $\hat{R}(s)$. In addition, the maximum semiparametric likelihood estimator $\tilde{A}$ of the area A under the ROC curve is equal to 0·855. An approximate 95% semiparametric confidence interval for A is given by (0·789, 0·911).

Example 2. Venkatraman & Begg (1996) discussed a melanoma dataset in which the definitive diagnosis of a pigmented lesion suspected of being a melanoma involves a biopsy.

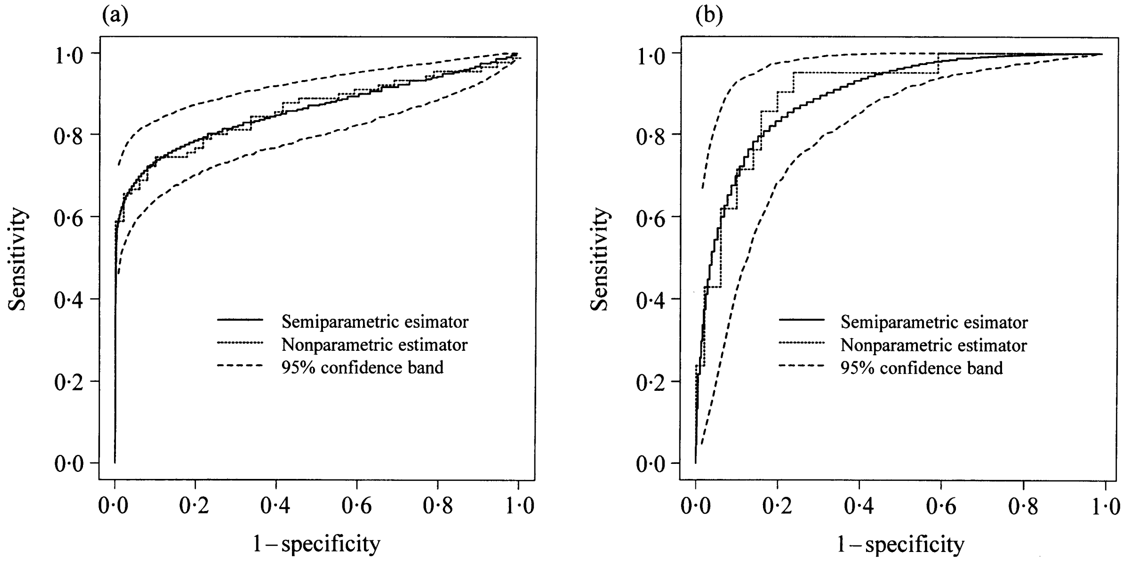

Fig. 1: Examples 1 and 2. Semiparametric and nonparametric estimators of the ROC curve based on (a) pancreatic CA19-9 data and (b) melanoma data.

There are two systems that can be used to evaluate suspicious lesions. One is the clinical score system given by doctors, and the other system is dermoscope. Here we use data only from the clinical score system. The Kolmogorov–Smirnov-type statistic of Qin & Zhang (1997) is used again for testing the validity of the logistic regression model (1·1) with $r(x)=x$, giving the observed p-value of 0·92 based on 5000 bootstrap replications. Therefore, model (1·1) with $r(x)=x$ is a good choice for the melanoma dataset. Figure 1(b) shows the semiparametric estimator $\tilde{R}(s)$ and the nonparametric estimator $\hat{R}(s)$ together with the 95% bootstrap confidence band. Again, $\tilde{R}(s)$ is much smoother than $\hat{R}(s)$. Moreover, an approximate 95% semiparametric confidence interval for A is given by $(0{\cdot}811, 0{\cdot}967)$ with $\tilde{A}=0{\cdot}889$.

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Appendix

Proofs

Proof of Theorem 1. Write

$$A_0(t)=\int_{-\infty}^{t}\frac{\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}}{1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}}\,dG(t),\quad A_0=A_0(\infty),$$

$$A_1(t)=\int_{-\infty}^{t}\frac{\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}}{1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}}\,r(t)\,dG(t),\quad A_1=A_1(\infty),$$

$$A_2=\int\frac{\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}}{1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}}\,r(t)\{r(t)\}^{\mathrm{T}}\,dG(t),\quad A=\begin{pmatrix}A_0 & A_1^{\mathrm{T}}\\ A_1 & A_2\end{pmatrix},\quad S=\frac{\rho}{1+\rho}A,$$

$$H_1(t)=\frac{1}{n_0}\sum_{i=1}^{n}\frac{I(T_i\leqslant t)}{1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(T_i)\}},\quad H_2(t)=\frac{\rho}{n}(A_0(t),A_1^{\mathrm{T}}(t))S^{-1}\begin{pmatrix}\dfrac{\partial l(\alpha_0,\beta_0)}{\partial\alpha}\\ \dfrac{\partial l(\alpha_0,\beta_0)}{\partial\beta}\end{pmatrix}. \tag{A·1}$$

Moreover, let

$$Q_n(t)=-[U_1(t)-\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}U_2(t)-\rho^{-1}U_3(t)]$$

with

$$U_1(t)=\hat{F}(t)-F(t),\quad U_2(t)=H_1(t)-H_2(t)-G(t),\quad U_3(t)=H_1(t)-\hat{G}(t)-H_2(t).$$

It can be shown after some albegra that

$$\begin{aligned}-\{\tilde{R}(1-s)-R(1-s)\}&=\tilde{F}\{\tilde{G}^{-1}(s)\}-F\{G^{-1}(s)\}\\&=U_1\{G^{-1}(s)\}+F\{\tilde{G}^{-1}(s)\}-F\{G^{-1}(s)\}+U_1\{\tilde{G}^{-1}(s)\}-U_1\{G^{-1}(s)\}\\&\quad-\frac{1}{\rho}(\tilde{G}\{G^{-1}(s)\}-\hat{G}\{G^{-1}(s)\}+U_4\{\tilde{G}^{-1}(s)\}-U_4\{G^{-1}(s)\}\\&\quad-[U_5\{\tilde{G}^{-1}(s)\}-U_5\{G^{-1}(s)\}]),\end{aligned} \tag{A·2}$$

where $U_4(t)=\tilde{G}(t)-G(t)$ and $U_5(t)=\hat{G}(t)-G(t)$. A first-order Taylor expansion and Theorem 4.1 of Zhang (2000) give

$$F\{\tilde{G}^{-1}(s)\}-F\{G^{-1}(s)\}=-\exp[\alpha_0+\beta_0^{\mathrm{T}}r\{G^{-1}(s)\}]U_2\{G^{-1}(s)\}+R_{1n}\{G^{-1}(s)\}, \tag{A·3}$$

where $\sup_{a\leqslant s\leqslant b}|R_{1n}\{G^{-1}(s)\}|=o_p(n^{-1/2})$. Furthermore, Theorem 2.1 of Zhang (2000) implies that

$$\tilde{G}\{G^{-1}(s)\}-\hat{G}\{G^{-1}(s)\}=U_3\{G^{-1}(s)\}+R_{2n}\{G^{-1}(s)\}, \tag{A·4}$$

where $\sup_{a\leqslant s\leqslant b}|R_{2n}\{G^{-1}(s)\}|=o_p(n^{-1/2})$. Moreover,

$$\phi_n=\sup_{a\leqslant s\leqslant b}|\tilde{G}^{-1}(s)-G^{-1}(s)|=O_p(n^{-1/2})$$

by part (b) of Lemma 6.1 of Zhang (2000). It now follows from part (d) of Lemma 6.1 of Zhang (2000) that

$$\begin{aligned}\sup_{a\leqslant s\leqslant b}|U_4\{\tilde{G}^{-1}(s)\}-U_4\{G^{-1}(s)\}|&=\sup_{a\leqslant s\leqslant b}|\tilde{G}\{\tilde{G}^{-1}(s)\}-G\{\tilde{G}^{-1}(s)\}-[\tilde{G}\{G^{-1}(s)\}-G\{G^{-1}(s)\}]|\\&\leqslant\sup_{G^{-1}(a)\leqslant x\leqslant G^{-1}(b)}\ \sup_{|y-x|\leqslant\phi_n}|\tilde{G}(x)-G(x)-\{\tilde{G}(y)-G(y)\}|\\&=o_p(n^{-1/2}).\end{aligned} \tag{A·5}$$

In addition, it follows from Lemma 1 of Bahadur (1966) and part (iv) of Lemma 8.5 of Zhang (1997) that

$$\begin{aligned}\sup_{a\leqslant s\leqslant b}|U_1\{\tilde{G}^{-1}(s)\}-U_1\{G^{-1}(s)\}|&=\sup_{a\leqslant s\leqslant b}|\hat{F}\{\tilde{G}^{-1}(s)\}-F\{\tilde{G}^{-1}(s)\}-[\hat{F}\{G^{-1}(s)\}-F\{G^{-1}(s)\}]|\\&\leqslant\sup_{G^{-1}(a)\leqslant x\leqslant G^{-1}(b)}\ \sup_{|y-x|\leqslant\phi_n}|\hat{F}(x)-F(x)-\{\hat{F}(y)-F(y)\}|\\&=o_p(n^{-1/2}),\end{aligned} \tag{A·6}$$

$$\begin{aligned}\sup_{a\leqslant s\leqslant b}|U_5\{\tilde{G}^{-1}(s)\}-U_5\{G^{-1}(s)\}|&=\sup_{a\leqslant s\leqslant b}|\hat{G}\{\tilde{G}^{-1}(s)\}-G\{\tilde{G}^{-1}(s)\}-[\hat{G}\{G^{-1}(s)\}-G\{G^{-1}(s)\}]|\\&\leqslant\sup_{G^{-1}(a)\leqslant x\leqslant G^{-1}(b)}\ \sup_{|y-x|\leqslant\phi_n}|\hat{G}(x)-G(x)-\{\hat{G}(y)-G(y)\}|\\&=o_p(n^{-1/2}).\end{aligned} \tag{A·7}$$

The combination of (A·2)–(A·7) yields

$$\tilde{R}(s)-R(s)=Q_n\{G^{-1}(1-s)\}+o_p(n^{-1/2}). \tag{A·8}$$

To prove weak convergence of $\sqrt{n}\{\tilde{R}(s)-R(s)\}$, it suffices to show by (A·8) that

$$\sqrt{n}Q_n(G^{-1})\rightarrow W(G^{-1})$$

weakly in $\mathscr{D}[1-b, 1-a]$. It is seen that $E\{Q_n(t)\}=0$ since $E\{U_1(t)\}=E\{U_2(t)\}=E\{U_3(t)\}=0$. Moreover, it can be shown after very extensive algebra that

$$\operatorname{cov}\{U_1(s), U_2(t)\}=\operatorname{cov}\{U_1(s), U_3(t)\}, \quad \operatorname{cov}\{U_2(s), U_3(t)\}=0,$$

$$\begin{aligned}
&\operatorname{cov}\{\sqrt{n}Q_n(s), \sqrt{n}Q_n(t)\}\\
&\quad=n\left[\operatorname{cov}\{U_1(s), U_1(t)\}+\frac{f(s)}{g(s)}\frac{f(t)}{g(t)}\operatorname{cov}\{U_2(s), U_2(t)\}+\frac{1}{\rho^2}\operatorname{cov}\{U_3(s), U_3(t)\}\right.\\
&\qquad\left.-\left\{\frac{f(t)}{g(t)}+\frac{1}{\rho}\right\}\operatorname{cov}\{U_1(s), U_2(t)\}-\left\{\frac{f(s)}{g(s)}+\frac{1}{\rho}\right\}\operatorname{cov}\{U_2(s), U_1(t)\}\right]\\
&\quad=\frac{1+\rho}{\rho}\{F(s\wedge t)-F(s)F(t)\}\\
&\qquad-\frac{f(s)}{g(s)}\frac{f(t)}{g(t)}(1-\rho)\left\{G(s\wedge t)-G(s)G(t)-\rho A_0(s\wedge t)+\rho(A_0(s), A_1^{\mathrm{T}}(s))A^{-1}\begin{pmatrix}A_0(t)\\A_1(t)\end{pmatrix}\right\}\\
&\qquad+\frac{1}{\rho^2}\rho(1+\rho)\left\{A_0(s\wedge t)-(A_0(s), A_1^{\mathrm{T}}(s))A^{-1}\begin{pmatrix}A_0(t)\\A_1(t)\end{pmatrix}\right\}\\
&\qquad-\left\{\frac{f(t)}{g(t)}+\frac{1}{\rho}\right\}(1+\rho)\left\{A_0(s\wedge t)-(A_0(s), A_1^{\mathrm{T}}(s))A^{-1}\begin{pmatrix}A_0(t)\\A_1(t)\end{pmatrix}\right\}\\
&\qquad-\left\{\frac{f(s)}{g(s)}+\frac{1}{\rho}\right\}(1+\rho)\left\{A_0(s\wedge t)-(A_0(s), A_1^{\mathrm{T}}(s))A^{-1}\begin{pmatrix}A_0(t)\\A_1(t)\end{pmatrix}\right\}\\
&\quad=\frac{1+\rho}{\rho}\{F(s\wedge t)-F(s)F(t)\}+(1+\rho)\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(s)\}\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}\{G(s\wedge t)-G(s)G(t)\}\\
&\qquad-\frac{1+\rho}{\rho}[1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(s)\}][1+\rho\exp\{\alpha_0+\beta_0^{\mathrm{T}}r(t)\}]\\
&\qquad\times\left\{A_0(s\wedge t)-(A_0(s), A_1^{\mathrm{T}}(s))A^{-1}\begin{pmatrix}A_0(t)\\A_1(t)\end{pmatrix}\right\}\\
&\quad=K(s,t)=E\{W(s)W(t)\}.
\end{aligned}$$

This, along with the central limit theorem for sample means and the Cramer–Wold device, implies that the finite-dimensional distributions of $\sqrt{n}Q_n$ converge weakly to those of W. By employing the tightness criteria in Billingsley (1968, Ch. 3), we can show that the process $\{\sqrt{n}Q_n(t), -\infty\leqslant t\leqslant\infty\}$ is tight in $\mathscr{D}[-\infty, \infty]$. As a result, we have $\sqrt{n}Q_n(G^{-1})\rightarrow W(G^{-1})$ weakly in $\mathscr{D}[1-b, 1-a]$. This completes the proof. □

Proof of Theorem 2. It follows from Theorem 1 and the Continuous Mapping Theorem (Billingsley, 1968, p. 30) that

$$\sqrt{n}(\tilde{A}-A)=\int_0^1\sqrt{n}\{\tilde{R}(s)-R(s)\}\,ds\rightarrow\int_0^1 W\{G^{-1}(1-s)\}\,ds\sim N\{0, \sigma^2(\alpha_0, \beta_0, G)\}$$

in distribution, where

$$\sigma^2(\alpha_0, \beta_0, G) = \int_0^1 \int_0^1 K\{G^{-1}(s), G^{-1}(t)\}\, ds\, dt = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K(s, t)\, dG(s)\, dG(t).$$

The proof is complete. □

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